

CBSE • SESSION 2026–27

Class 12 Mathematics

Standard Mathematics — Code 041



Complete syllabus, unit-wise weightage, question-paper design, formula sheets for all 13 chapters, and step-by-step worked examples with CBSE-style board-exam mark allocation.

Syllabus & weightage: Official CBSE Curriculum for 2026–27
(cbseacademic.nic.in)

Paper design & marking style: Official CBSE Sample Question Paper & Marking Scheme, 2025–26 (the latest released at the time of writing — the 2026–27 SQP had not yet been issued)

Compiled 30 August 2026

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1. Syllabus at a Glance

COURSE STRUCTURE & UNIT-WISE WEIGHTAGE

Unit	Title	Marks
I	Relations and Functions	8
II	Algebra	10
III	Calculus	35
IV	Vectors and Three-Dimensional Geometry	14
V	Linear Programming	5
VI	Probability	8
Theory Total		80
Internal Assessment		20
Grand Total		100

Official note: "No chapter-wise weightage is assigned. Care should be taken to cover all the chapters as specified in the curriculum." Calculus carries the largest share of marks ($35/80 \approx 44\%$), so it deserves the most practice time, but every chapter can still be examined.

PRESCRIBED NCERT TEXTBOOKS

- Mathematics, Textbook for Class XII, Parts I & II — NCERT
- Mathematics Exemplar Problems, Class XII — NCERT
- Mathematics Lab Manual, Class XII — NCERT

COGNITIVE-LEVEL (DIFFICULTY) WEIGHTAGE

Competency	Marks	%
Remembering & Understanding	44	55%
Applying	20	25%
Analysing / Evaluating / Creating	16	20%

No overall choice is given in the paper, but 33% internal choice is provided within every section.

2. Question Paper Design (Blueprint)

The exact 2026–27 Sample Question Paper had not been released by CBSE as of late August 2026 (SQPs are usually issued around September–October for the year's board exam). The table below is the official design used in the most recently released Sample Question Paper (2025–26), which has been the stable CBSE Maths pattern for several years and is the best available guide to how the 2027 board paper will be structured. Time allowed: **3 hours**; maximum marks: **80**; total **38 questions**, all compulsory; calculators are **not** permitted.

Section	Question Nos.	Type	Marks each	Count	Section Total
A	1–18	Multiple Choice Questions	1	18	18
A	19–20	Assertion–Reason	1	2	2
B	21–25	Very Short Answer (VSA)	2	5	10
C	26–31	Short Answer (SA)	3	6	18
D	32–35	Long Answer (LA)	5	4	20
E	36–38	Case-Study Based (with sub-parts)	4	3	12
Total					80

Source: CBSE Sample Question Paper, Mathematics (Code 041), Class XII, 2025–26 — cbseacademic.nic.in.

Internal choice pattern: Section B — 2 of 5 questions carry a choice; Section C — 3 of 6; Section D — 2 of 4; Section E — one sub-part in 2 of the 3 case studies. This 33% internal-choice rule applies to every section.

3. How CBSE Awards Marks — Step-Marking Principles

CBSE does not grade a Maths answer only on the final result. The official Marking Scheme released alongside each Sample/Board paper breaks every multi-mark question into small steps and assigns a fraction of the total (often $\frac{1}{2}$ mark) to each step. Reading the worked examples in Sections 5–10 of this document (each carries its own step/marks table) makes the pattern visible; the general principles behind it are:

- **Every valid step earns marks, independent of the final answer** — setting up the correct equation, formula or matrix is credited even if a later arithmetic slip changes the final number.
- **Marks are typically split as:** recognising the right concept/formula and setting up the problem ($\approx 20\text{--}30\%$ of the question), the main derivation/computation ($\approx 40\text{--}60\%$), and stating the final answer correctly, with correct units/interpretation where relevant ($\approx 10\text{--}20\%$).
- **Follow-through credit:** if an early step has a small error but the student correctly applies the method to their (wrong) intermediate result, later steps still earn marks.

- **Alternative correct methods** that are mathematically valid are awarded full marks, even if they differ from the method shown in the official scheme.
- **Case-study (Section E) sub-parts** are marked independently — 1 mark sub-parts require only the final justified answer, while 2-mark sub-parts expect a short working.
- **No marks are deducted** for using a different (valid) variable name, for not re-stating "Hence Proved," or for leaving an answer as an unsimplified but correct fraction/surd, unless a question explicitly asks for simplification.

Illustrated using four complete solutions reproduced from the official CBSE Marking Scheme, Mathematics (041), Class XII, 2025–26 (see Algebra, Calculus and Probability worked examples ahead, each marked "Official CBSE Marking Scheme").

4. Internal Assessment (20 Marks)

Component	Marks	Detail
Periodic Tests (best 2 of 3)	10	Three tests held tentatively in Jul–Aug, Nov, and Dec–Jan; the average of the best two is scaled to 10.
Mathematics Activities	10	5 marks — performance & record throughout the year; 3 marks — year-end activity/practical test; 2 marks — viva-voce.

Unit I · Relations and Functions

8 marks | 2 chapters | ~30 periods

5(a). Relations and Functions

FORMULAS & KEY DEFINITIONS

Relation: a relation R from set A to B is a subset of $A \times B$.

Reflexive: $(a,a) \in R \forall a \in A$

Symmetric: $(a,b) \in R \Rightarrow (b,a) \in R$

Transitive: $(a,b) \in R$ and $(b,c) \in R \Rightarrow (a,c) \in R$

Equivalence relation: reflexive + symmetric + transitive

One-one (injective): $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$

Onto (surjective): range of f = co-domain of f

Bijjective: one-one and onto ($\Rightarrow f$ is invertible)

Composition: $(g \circ f)(x) = g(f(x))$

WORKED EXAMPLE

Show that a Relation is an Equivalence Relation

5 marks · typical LA style

Q. Let $A = \{1, 2, 3, 4, 5\}$. Show that the relation $R = \{(a, b) : |a - b| \text{ is even}\}$ is an equivalence relation.

Step	Working	Marks
1	Reflexive: $ a - a = 0$, which is even, for every $a \in A$. So $(a, a) \in R \forall a$. R is reflexive.	1
2	Symmetric: If $(a, b) \in R$ then $ a - b $ is even. Since $ a - b = b - a $, $ b - a $ is also even, so $(b, a) \in R$. R is symmetric.	1
3	Transitive: If $(a, b) \in R$ and $(b, c) \in R$, then $ a - b $ and $ b - c $ are both even, so a, b have the same parity and b, c have the same parity. Hence a, c have the same parity, so $ a - c $ is even and $(a, c) \in R$.	2
4	Since R is reflexive, symmetric and transitive, R is an equivalence relation. (Its equivalence classes are $\{1, 3, 5\}$ and $\{2, 4\}$.)	1
Total		5

5(b). Inverse Trigonometric Functions

DOMAIN & RANGE (PRINCIPAL VALUE BRANCHES)

Function	Domain	Range (Principal Branch)
$\sin^{-1}x$	$[-1, 1]$	$[-\pi/2, \pi/2]$
$\cos^{-1}x$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1}x$	$\mathbb{R}$	$(-\pi/2, \pi/2)$
$\cot^{-1}x$	$\mathbb{R}$	$(0, \pi)$
$\sec^{-1}x$	$\mathbb{R} - (-1, 1)$	$[0, \pi] - \{\pi/2\}$
$\operatorname{cosec}^{-1}x$	$\mathbb{R} - (-1, 1)$	$[-\pi/2, \pi/2] - \{0\}$

KEY IDENTITIES

$$\sin^{-1}x + \cos^{-1}x = \pi/2 \quad \tan^{-1}x + \cot^{-1}x = \pi/2 \quad \sec^{-1}x + \operatorname{cosec}^{-1}x = \pi/2$$

$$\sin^{-1}(-x) = -\sin^{-1}x, \quad \cos^{-1}(-x) = \pi - \cos^{-1}x, \quad \tan^{-1}(-x) = -\tan^{-1}x$$

$$\tan^{-1}x + \tan^{-1}y = \tan^{-1}(x + y)/(1 - xy), \quad \text{when } xy < 1$$

$$\tan^{-1}x - \tan^{-1}y = \tan^{-1}(x - y)/(1 + xy)$$

$$2\tan^{-1}x = \sin^{-1} 2x/(1+x^2) = \cos^{-1} (1-x^2)/(1+x^2) = \tan^{-1} 2x/(1-x^2)$$

WORKED EXAMPLE

Prove an Inverse-Trig Identity

5 marks

Q. Prove that $\tan^{-1}(1/5) + \tan^{-1}(1/7) + \tan^{-1}(1/3) + \tan^{-1}(1/8) = \pi/4$.

Step	Working	Marks
1	Group into pairs: $[\tan^{-1}(1/5) + \tan^{-1}(1/7)] + [\tan^{-1}(1/3) + \tan^{-1}(1/8)]$	1
2	Apply $\tan^{-1}x + \tan^{-1}y = \tan^{-1}[(x+y)/(1-xy)]$ to each pair: Pair 1 = $\tan^{-1}[(1/5+1/7)/(1-1/35)] = \tan^{-1}(12/34) = \tan^{-1}(6/17)$ Pair 2 = $\tan^{-1}[(1/3+1/8)/(1-1/24)] = \tan^{-1}(11/23)$	1.5
3	Combine again: $\tan^{-1}(6/17) + \tan^{-1}(11/23) = \tan^{-1}[(6/17+11/23)/(1-(6/17)(11/23))] = \tan^{-1}(325/391 \div 325/391) = \tan^{-1}(1)$	1.5
4	$\tan^{-1}(1) = \pi/4$. Hence proved.	1
Total		5

Unit II · Algebra

10 marks | 2 chapters | ~50 periods

6(a). Matrices

FORMULAS

Order: a matrix with m rows and n columns has order $m \times n$.

Equality: same order and every corresponding entry equal.

Operations: $A+B$, kA (scalar multiplication), AB defined only when columns of A = rows of B .

Properties: $A(BC) = (AB)C$, $A(B+C) = AB+AC$, $AB \neq BA$ in general.

Transpose: $(A')' = A$, $(A+B)' = A'+B'$, $(AB)' = B'A'$, $(kA)' = kA'$

Symmetric: $A' = A$ **Skew-symmetric:** $A' = -A$

Any square matrix $A = \frac{1}{2}(A+A') + \frac{1}{2}(A-A')$ (symmetric part + skew-symmetric part)

Invertible matrix: A^{-1} exists $\Leftrightarrow |A| \neq 0$; $(AB)^{-1} = B^{-1}A^{-1}$

WORKED EXAMPLE

Express a Matrix as Symmetric + Skew-Symmetric

3 marks

Q. Express $A = \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix}$ as the sum of a symmetric and a skew-symmetric matrix.

Step	Working	Marks
1	$A' = \begin{bmatrix} 3 & 1 \\ 5 & -1 \end{bmatrix}$	0.5
2	Symmetric part $P = \frac{1}{2}(A+A') = \frac{1}{2} \begin{bmatrix} 6 & 6 \\ 6 & -2 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & -1 \end{bmatrix}$	1
3	Skew-symmetric part $Q = \frac{1}{2}(A-A') = \frac{1}{2} \begin{bmatrix} 0 & 4 \\ -4 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$	1
4	Check: $P + Q = \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix} = A$ ✓	0.5
Total		3

6(b). Determinants

FORMULAS

$$2 \times 2: |[a,b],[c,d]| = ad - bc$$

Minor M_{ij} = determinant left after deleting row i , column j . **Cofactor** $A_{ij} = (-1)^{i+j}M_{ij}$

Area of triangle with vertices (x_1, y_1) , (x_2, y_2) , (x_3, y_3) : $\frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$. Points are collinear when this area = 0.

adj(A) = transpose of the cofactor matrix of A . $A \cdot \text{adj}(A) = \text{adj}(A) \cdot A = |A| I$

Inverse: $A^{-1} = (1/|A|) \cdot \text{adj}(A)$, valid only if $|A| \neq 0$

Matrix method for $AX = B$: $X = A^{-1}B$. If $|A| \neq 0 \Rightarrow$ unique solution. If $|A| = 0$: infinitely many solutions when $(\text{adj } A)B = O$, otherwise no solution.

WORKED EXAMPLE — OFFICIAL CBSE MARKING SCHEME

Solve a System of Equations by the Matrix Method

5 marks · Official CBSE Marking Scheme, 2025–26, Q32

Q. Given $A = [[3, -6, -1], [2, -5, -1], [-2, 4, 1]]$ and $B = [[1, -2, -1], [0, -1, -1], [2, 0, 3]]$, first show $AB = I$, and hence solve the system: $3x - 6y - z = 3$, $2x - 5y - z = -2$, $-2x + 4y + z = 5$.

Step	Working	Marks
1	Multiply out AB and show $AB = I$ (so $B = A^{-1}$)	1
2	Write the system in matrix form $AX = C$, where $X = [x, y, z]'$ and $C = [3, -2, 5]'$	0.5
3	Since $ A = -1 \neq 0$, the system has a unique solution	0.5
4	$X = A^{-1}C = BC$	0.5
5	$BC = [[1, -2, -1], [0, -1, -1], [2, 0, 3]] \cdot [3, -2, 5]' = [2, -3, 21]'$	0.5
6	$x = 2, y = -3, z = 21$	1
Total		5

Reproduced (with the coefficient matrix reconstructed from the official solution) from the CBSE Marking Scheme, Mathematics 041, Class XII, 2025–26 — cbseacademic.nic.in.

Unit III · Calculus

35 marks | 5 chapters | ~80 periods — the highest-weightage unit

7(a). Continuity and Differentiability

FORMULAS

f is **continuous** at $x=a$ if $\lim_{x \rightarrow a} f(x) = f(a)$. f is **differentiable** at $x=a$ if LHD = RHD there. Every differentiable function is continuous (converse is false).

Chain rule: $dy/dx = (dy/du) \cdot (du/dx)$

$$d/dx(\sin^{-1}x) = 1/\sqrt{1-x^2} \quad d/dx(\cos^{-1}x) = -1/\sqrt{1-x^2} \quad d/dx(\tan^{-1}x) = 1/(1+x^2)$$

$$d/dx(\cot^{-1}x) = -1/(1+x^2) \quad d/dx(\sec^{-1}x) = 1/(x\sqrt{x^2-1}) \quad d/dx(\operatorname{cosec}^{-1}x) = -1/(x\sqrt{x^2-1})$$

$$d/dx(e^x) = e^x \quad d/dx(a^x) = a^x \ln a \quad d/dx(\ln x) = 1/x$$

Logarithmic differentiation for $y = u^v$: take log on both sides, then differentiate implicitly.

Parametric form: if $x=f(t)$, $y=g(t)$, then $dy/dx = (dy/dt) \div (dx/dt)$

WORKED EXAMPLE

Logarithmic Differentiation

5 marks · typical LA style

Q. If $y = (\sin x)^x + \sin^{-1}\sqrt{x}$, find dy/dx .

Step	Working	Marks
1	Let $u = (\sin x)^x$. Take log: $\log u = x \cdot \log(\sin x)$	1
2	Differentiate implicitly w.r.t. x : $(1/u)(du/dx) = \log(\sin x) + x \cdot \cot x$	1
3	$du/dx = (\sin x)^x [\log(\sin x) + x \cot x]$	1
4	Let $v = \sin^{-1}\sqrt{x}$. $dv/dx = 1/\sqrt{1-x} \cdot 1/(2\sqrt{x}) = 1/(2\sqrt{x-x^2})$	1
5	$dy/dx = du/dx + dv/dx = (\sin x)^x [\log(\sin x) + x \cot x] + 1/(2\sqrt{x-x^2})$	1
Total		5

7(b). Application of Derivatives

FORMULAS

Rate of change: dy/dx (or ds/dt) gives the instantaneous rate.

f is **increasing** on an interval if $f'(x) > 0$ there; **decreasing** if $f'(x) < 0$.

Tangent at (x_1, y_1) : $y - y_1 = f'(x_1)(x - x_1)$. **Normal:** $y - y_1 = -1/f'(x_1) \cdot (x - x_1)$

Approximation: $f(x + \Delta x) \approx f(x) + f'(x) \cdot \Delta x$

Critical points: $f'(x) = 0$. **First derivative test:** sign change of f' around the point. **Second derivative test:** $f''(x) < 0 \Rightarrow$ local max; $f''(x) > 0 \Rightarrow$ local min.

Absolute max/min on $[a, b]$: compare f at all critical points inside (a, b) and at the endpoints a, b .

WORKED EXAMPLE

Optimisation (Maxima/Minima)

5 marks · typical LA style

Q. An open box is to be made from a square sheet of side 24 cm by cutting equal squares of side x from each corner and folding up the sides. Find x for which the volume is maximum, and find that maximum volume.

Step	Working	Marks
1	Volume $V(x) = x(24-2x)^2$, $0 < x < 12$	1
2	$dV/dx = (24-2x)^2 + x \cdot 2(24-2x)(-2) = (24-2x)(24-6x)$	1
3	Set $dV/dx = 0 \Rightarrow x = 12$ (rejected, gives $V=0$) or $x = 4$	1
4	d^2V/dx^2 at $x=4$ is negative $\Rightarrow V$ is maximum at $x = 4$	1
5	$V_{\max} = 4(24-8)^2 = 4 \times 256 = 1024 \text{ cm}^3$ at $x = 4 \text{ cm}$	1
Total		5

7(c). Integrals

STANDARD INTEGRALS

$$\int x^n dx = x^{n+1}/(n+1) + C \quad \int (1/x) dx = \ln|x| + C \quad \int e^x dx = e^x + C \quad \int a^x dx = a^x/\ln a + C$$

$$\int \sin x dx = -\cos x + C \quad \int \cos x dx = \sin x + C \quad \int \sec^2 x dx = \tan x + C \quad \int \operatorname{cosec}^2 x dx = -\cot x + C$$

$$\int \sec x \cdot \tan x dx = \sec x + C \quad \int \tan x dx = \ln|\sec x| + C \quad \int \sec x dx = \ln|\sec x + \tan x| + C$$

$$\int dx/\sqrt{a^2-x^2} = \sin^{-1}(x/a) + C \quad \int dx/(a^2+x^2) = (1/a)\tan^{-1}(x/a) + C$$

$$\int dx/(a^2-x^2) = (1/2a)\ln|(a+x)/(a-x)| + C \quad \int dx/\sqrt{x^2 \pm a^2} = \ln|x + \sqrt{x^2 \pm a^2}| + C$$

$$\int \sqrt{a^2-x^2} dx = (x/2)\sqrt{a^2-x^2} + (a^2/2)\sin^{-1}(x/a) + C$$

Integration by parts: $\int u \cdot v dx = u \int v dx - \int [u' \cdot \int v dx] dx$ (choose u using ILATE)

$$\int e^x [f(x) + f'(x)] dx = e^x f(x) + C$$

Definite integral properties: $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$; $\int_0^a f(x) dx = \int_0^a f(a-x) dx$; $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$ if f is even, 0 if f is odd.

WORKED EXAMPLE — OFFICIAL CBSE MARKING SCHEME

Definite Integral Using a Property

5 marks · Official CBSE Marking Scheme, 2025–26, Q33

Q. Evaluate $\int_0^{\pi/4} \log(1+\tan x) dx$.

Step	Working	Marks
1	Let $I = \int_0^{\pi/4} \log(1+\tan x) dx$	0.5
2	Apply $\int_0^a f(x) dx = \int_0^a f(a-x) dx$ with $a = \pi/4$: $I = \int_0^{\pi/4} \log(1+\tan(\pi/4-x)) dx$	1
3	Use $\tan(\pi/4-x) = (1-\tan x)/(1+\tan x)$, so $1+\tan(\pi/4-x) = 2/(1+\tan x)$	1
4	$I = \int_0^{\pi/4} [\log 2 - \log(1+\tan x)] dx = (\pi/4)\log 2 - I$	1
5	$2I = (\pi/4)\log 2$	1
6	$I = (\pi/8) \log 2$	0.5
Total		5

Reproduced from the CBSE Marking Scheme, Mathematics 041, Class XII, 2025–26, Q33 (solved there via the substitution $x = \tan \theta$, equivalent to the direct form shown) — cbseacademic.nic.in.

7(d). Application of Integrals

FORMULAS

Area under a curve $y=f(x)$ from $x=a$ to $x=b$ ($f(x) \geq 0$): $A = \int_a^b y \, dx$

Area between two curves: $A = \int_a^b [f(x) - g(x)] \, dx$, where $f(x) \geq g(x)$ on $[a, b]$

Area w.r.t. y-axis: $A = \int_c^d x \, dy$

Circle $x^2 + y^2 = a^2$: full area = πa^2 **Ellipse** $x^2/a^2 + y^2/b^2 = 1$: full area = πab

WORKED EXAMPLE

Area Bounded by a Parabola and a Line

5 marks · typical LA style

Q. Find the area of the region bounded by the parabola $y^2 = 4ax$ and its latus rectum $x = a$.

Step	Working	Marks
1	By symmetry about the x-axis, required area $A = 2 \int_0^a y \, dx$	1
2	From $y^2 = 4ax$, $y = 2\sqrt{ax}$ for the upper half	1
3	$A = 2 \int_0^a 2\sqrt{ax} \, dx = 4\sqrt{a} \int_0^a \sqrt{x} \, dx$	1.5
4	$= 4\sqrt{a} \cdot \left[\frac{2}{3} x^{3/2} \right]_0^a = 4\sqrt{a} \cdot \frac{2}{3} a^{3/2}$	1
5	$A = 8a^2/3$ sq. units	0.5
Total		5

7(e). Differential Equations

FORMULAS

Order = highest derivative present; **degree** = power of the highest derivative (after removing radicals/fractions in derivatives).

Variable separable: $dy/dx = f(x)g(y) \Rightarrow \int dy/g(y) = \int f(x)dx$

Homogeneous DE: $dy/dx = F(y/x)$; substitute $y = vx$, then $dy/dx = v + x(dv/dx)$

Linear DE in y: $dy/dx + Py = Q$ (P, Q functions of x). Integrating factor $IF = e^{\int P dx}$. Solution: $y \cdot IF = \int Q \cdot IF dx + C$

Linear DE in x: $dx/dy + P_1x = Q_1$ (P_1, Q_1 functions of y). $IF = e^{\int P_1 dy}$; solution $x \cdot IF = \int Q_1 \cdot IF dy + C$

WORKED EXAMPLE

Linear Differential Equation

5 marks · typical LA style

Q. Solve: $x(dy/dx) + 2y = x^2 \log x$ ($x > 0$).

Step	Working	Marks
1	Divide by x to get standard linear form: $dy/dx + (2/x)y = x \log x$. Here $P = 2/x$, $Q = x \log x$	1
2	$IF = e^{\int (2/x)dx} = e^{2 \ln x} = x^2$	1
3	Solution: $y \cdot x^2 = \int x^2 \cdot x \log x dx = \int x^3 \log x dx$	1
4	By parts ($u = \log x$, $dv = x^3 dx$): $\int x^3 \log x dx = (x^4/4) \log x - \int (x^4/4)(1/x)dx = (x^4/4) \log x - x^4/16$	1.5
5	$y \cdot x^2 = (x^4/4) \log x - x^4/16 + C$, i.e. $y = (x^2/4) \log x - x^2/16 + C/x^2$	0.5
Total		5

Unit IV · Vectors and Three-Dimensional Geometry

14 marks | 2 chapters | ~30 periods

8(a). Vector Algebra

FORMULAS

Magnitude: $|\vec{a}| = \sqrt{x^2 + y^2 + z^2}$ **Direction cosines** l, m, n satisfy $l^2 + m^2 + n^2 = 1$ **Unit vector:** $\hat{a} = \vec{a}/|\vec{a}|$

Section formula (internal, ratio $m:n$): position vector = $(m\vec{b} + n\vec{a})/(m+n)$

Dot product: $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos\theta = a_1b_1 + a_2b_2 + a_3b_3$ **Projection of $\vec{a}$ on $\vec{b}$** = $(\vec{a} \cdot \vec{b})/|\vec{b}|$

Cross product: $\vec{a} \times \vec{b} = |\vec{a}||\vec{b}|\sin\theta \hat{n}$, computed as a 3×3 determinant with rows $\hat{i}, \hat{j}, \hat{k}$ / a_1, a_2, a_3 / b_1, b_2, b_3

Area of triangle = $\frac{1}{2}|\vec{a} \times \vec{b}|$ **Area of parallelogram** = $|\vec{a} \times \vec{b}|$

WORKED EXAMPLE

Unit Vector Perpendicular to Two Given Vectors

3 marks · typical SA style

Q. Find a unit vector perpendicular to both $\vec{a} = 2\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - 2\hat{k}$.

Step	Working	Marks
1	$\vec{a} \times \vec{b} = \det[\hat{i}, \hat{j}, \hat{k}; 2, 1, 2; 1, 2, -2] = \hat{i}(-2-4) - \hat{j}(-4-2) + \hat{k}(4-1) = -6\hat{i} + 6\hat{j} + 3\hat{k}$	1.5
2	$ \vec{a} \times \vec{b} = \sqrt{36+36+9} = \sqrt{81} = 9$	1
3	Unit vector = $(-6\hat{i} + 6\hat{j} + 3\hat{k})/9 = -(2/3)\hat{i} + (2/3)\hat{j} + (1/3)\hat{k}$	0.5
Total		3

8(b). Three-Dimensional Geometry

Note: the current CBSE syllabus restricts this chapter to **lines only** (planes were removed in the recent syllabus rationalisation).

FORMULAS

Direction ratios of the line joining (x_1, y_1, z_1) and (x_2, y_2, z_2) : $(x_2 - x_1, y_2 - y_1, z_2 - z_1)$

Cartesian equation of a line through (x_1, y_1, z_1) with direction ratios a, b, c : $(x - x_1)/a = (y - y_1)/b = (z - z_1)/c$ **Vector form**: $\vec{r} = \vec{a} + \lambda \vec{b}$

Angle between two lines with direction ratios (a_1, b_1, c_1) and (a_2, b_2, c_2) : $\cos \theta = |a_1 a_2 + b_1 b_2 + c_1 c_2| / [\sqrt{(a_1^2 + b_1^2 + c_1^2)} \cdot \sqrt{(a_2^2 + b_2^2 + c_2^2)}]$. Perpendicular when $a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$.

Shortest distance between skew lines $\vec{r} = \vec{a}_1 + \lambda \vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu \vec{b}_2$: $d = |(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)| / |\vec{b}_1 \times \vec{b}_2|$

WORKED EXAMPLE

Shortest Distance Between Two Skew Lines

5 marks · typical LA style

Q. Find the shortest distance between the lines $\vec{r} = (\hat{i} + 2\hat{j} + \hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k})$ and $\vec{r} = (2\hat{i} - \hat{j} - \hat{k}) + \mu(2\hat{i} + \hat{j} + 2\hat{k})$.

Step	Working	Marks
1	$\vec{a}_1 = (1, 2, 1)$, $\vec{b}_1 = (1, -1, 1)$; $\vec{a}_2 = (2, -1, -1)$, $\vec{b}_2 = (2, 1, 2)$. Compute $\vec{a}_2 - \vec{a}_1 = (1, -3, -2)$	1
2	$\vec{b}_1 \times \vec{b}_2 = \det[\hat{i}, \hat{j}, \hat{k}; 1, -1, 1; 2, 1, 2] = \hat{i}(-2-1) - \hat{j}(2-2) + \hat{k}(1+2) = (-3, 0, 3)$	1.5
3	$ \vec{b}_1 \times \vec{b}_2 = \sqrt{9+0+9} = \sqrt{18} = 3\sqrt{2}$	0.5
4	$(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = (1)(-3) + (-3)(0) + (-2)(3) = -9$	1
5	$d = -9 / 3\sqrt{2} = 3/\sqrt{2} = (3\sqrt{2})/2$ units	1
Total		5

Unit V · Linear Programming

5 marks | 1 chapter | ~20 periods

9. Linear Programming

FORMULAS & METHOD

Objective function: $Z = ax + by$, to be maximised or minimised, subject to linear **constraints** and non-negativity $x, y \geq 0$.

Feasible region: the set of all points satisfying every constraint simultaneously.

Corner Point Method: the optimal value of Z (if it exists) occurs at a corner (vertex) of the feasible region. Evaluate Z at every corner point and compare.

If the feasible region is **unbounded**, a maximum/minimum may or may not exist — it must be checked by testing whether the corresponding open half-plane has any common point with the feasible region.

WORKED EXAMPLE

Minimisation LPP (Corner Point Method)

5 marks · typical LA style

Q. Minimise $Z = 200x + 500y$ subject to $x + 2y \geq 10$, $3x + 4y \leq 24$, $x \geq 0$, $y \geq 0$.

Step	Working	Marks
1	Draw the lines $x+2y=10$ and $3x+4y=24$ and shade the feasible region satisfying both constraints and $x,y \geq 0$.	1
2	Find corner points: solving $x+2y=10$ and $3x+4y=24$ simultaneously gives $(4,3)$; $x+2y=10$ meets the y -axis at $(0,5)$; $3x+4y=24$ meets the y -axis at $(0,6)$. (Points on the x -axis are infeasible here.) So the feasible region is a triangle with vertices $(0,5)$, $(0,6)$, $(4,3)$.	2
3	Evaluate Z at each corner: $Z(0,5)=2500$, $Z(0,6)=3000$, $Z(4,3)=200(4)+500(3)=800+1500=2300$	1
4	Minimum $Z = 2300$ at $x=4$, $y=3$	1
Total		5

Unit VI · Probability

8 marks | 1 chapter | ~30 periods

10. Probability

FORMULAS

Conditional probability: $P(A|B) = P(A \cap B)/P(B)$, $P(B) > 0$

Multiplication theorem: $P(A \cap B) = P(A) \cdot P(B|A) = P(B) \cdot P(A|B)$

Independent events: $P(A \cap B) = P(A) \cdot P(B)$

Total probability theorem: if $E_1, E_2, \dots, E_n$ partition the sample space, $P(A) = \sum P(E_i) \cdot P(A|E_i)$

Bayes' theorem: $P(E_i|A) = [P(E_i) \cdot P(A|E_i)] / \sum_j [P(E_j) \cdot P(A|E_j)]$

WORKED EXAMPLE — OFFICIAL CBSE MARKING SCHEME

Bayes' Theorem

5 marks · Official CBSE Marking Scheme, 2025–26, Q38

Q. In a survey, 60% of students use screens for more than 4 hrs/day (E_1), 30% for 2–4 hrs (E_2), and 10% for less than 2 hrs (E_3). The probability of showing anxiety symptoms (A) is 80% for E_1 , 70% for E_2 , and 30% for E_3 . If a randomly chosen student shows anxiety symptoms, find the probability that the student uses screens for more than 4 hrs/day.

Step	Working	Marks
1	Define events E_1, E_2, E_3 (screen-time groups) and A (shows anxiety symptoms)	1
2	$P(E_1)=60/100$, $P(E_2)=30/100$, $P(E_3)=10/100$	0.5
3	$P(A E_1)=80/100$, $P(A E_2)=70/100$, $P(A E_3)=30/100$	0.5
4	By the total probability theorem: $P(A) = (60/100)(80/100) + (30/100)(70/100) + (10/100)(30/100) = 72/100$	2
5	By Bayes' theorem: $P(E_1 A) = P(E_1)P(A E_1)/P(A) = (48/100)/(72/100) = 48/72 = 2/3$	2
Total		5

Reproduced from the CBSE Marking Scheme, Mathematics 041, Class XII, 2025–26, Q38 — cbseacademic.nic.in.

Sources

- CBSE Curriculum, Mathematics (Code 041), Class XII, 2026–27 — cbseacademic.nic.in/web_material/CurriculumMain27/SecPart2/Maths_SecP2_2026-27.pdf (course structure, unit weightage, detailed topics, question-paper design percentages, internal assessment, prescribed books)
- CBSE Sample Question Paper, Mathematics (Code 041), Class XII, 2025–26 — cbseacademic.nic.in/web_material/SQP/ClassXII_2025_26/Maths-SQP.pdf (section-wise blueprint; the 2026–27 SQP was not yet released as of August 2026)
- CBSE Marking Scheme, Mathematics (Code 041), Class XII, 2025–26 — cbseacademic.nic.in/web_material/SQP/ClassXII_2025_26/Maths-MS.pdf (step-wise mark allocation for Q32, Q33, Q38, reproduced above)
- NCERT Textbooks & Exemplar, Mathematics, Class XII, Parts I & II (formula content, standard results)

Compiled for personal study use from official CBSE sources and standard NCERT content, current as of 30 August 2026. Always cross-check against the final 2026–27 Sample Question Paper once CBSE releases it (typically September–October 2026).